

Twenty Years of Personality Computing: Threats, Challenges and Future Directions

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Personality Computing is a field at the intersection of Personality Psychology and Computer Science. Started in 2005, research in the field utilizes computational methods to understand and predict human personality traits. The expansion of the field has been very rapid and, by analyzing digital footprints (text, images, social media, etc.), it helped to develop systems that recognize and even replicate human personality. While offering promising applications in talent recruiting, marketing and healthcare, the ethical implications of Personality Computing are significant. Concerns include data privacy, algorithmic bias, and the potential for manipulation by personality-aware Artificial Intelligence. This paper provides an overview of the field, explores key methodologies, discusses the challenges and threats, and outlines potential future directions for responsible development and deployment of Personality Computing technologies.

CCS Concepts: • **Information systems** → **Clustering and classification**; *Personalization*; • **Computing methodologies** → *Information extraction*; *Supervised learning by classification*; • **Human-centered computing** → *Social tagging*; • **Security and privacy** → *Social aspects of security and privacy*.

Additional Key Words and Phrases: Personality Computing, Personality Recognition, Personality Perception, Personality Synthesis

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1 Introduction

Personality Computing is a research field started in 2005 that uses computational methods to understand and predict human personality. There are many digital footprints conveying information that computational models can decode and use in various ways to model behavior and interact with humans. These include text [176, 178, 179], images [87, 248, 249], video and audio tracks [132, 194, 222, 272], social media posts and preferences [158, 160, 246], credit card transactions [100, 266], mobile phone communication logs [53, 65, 196, 254], experience sampling apps [182], and physiological data [77]. The basis for training Personality Computing models comes from Personality Psychology [234], primarily using questionnaires to assess and quantify personality traits [37, 62]. The availability of data that combine digital footprints and scores from personality assessments allowed the development of systems that recognize human personality exploiting sophisticated computational techniques, including Natural Language Processing (NLP), Computer Vision (CV), Automatic Speech Recognition (ASR), Deep Learning (DL) and Social Network Analysis (SNA). Systems for Personality Computing have been applied in different domains, both at population (i.e: targeted marketing [46, 244], recommendation systems [68]) and individual level (i.e: mental health detection [298], job performance [15, 156], human-computer interaction [22, 23]).

Crucially, these algorithms were employed in numerous applications that pose serious threats to societal integrity, such as digital mass persuasion on voters in various elections [243, 295], surveillance and social scoring systems that integrate personality, raising serious concerns about the ethical use of Personality Computing. As a result, the data protection regulations (e.g., the European Union General Data Protection Regulation - GDPR¹) and the legislation on Artificial Intelligence (e.g., the European Union AI Act²) often limit Personality Computing. There are restrictions on the types of data that can be collected, rules prohibiting uses that could be discriminatory or harmful, and laws that require systems to be transparent about how personality inferences are made (i.e.: explanations for automated decisions involving personality). Recent advancements in Artificial Intelligence (AI) have led to Large Language Models (LLMs), which are able to detect and replicate human personality during interactions and role-playing tasks [121, 125, 126, 135, 136, 219, 275, 276], potentially reaching a high level of intimacy between AI systems and humans. This might yield to scenarios where AI systems can manipulate humans to engineer large scale social functions, such as voting and spending, with a disruptive impact on human societies [116]. This is why a survey of this field is essential to identify challenges and explore ways to mitigate the negative social impacts of Personality Computing.

The objective of this paper is to systematically examine the development of Personality Computing over the past two decades, with a focus on its methods, applications, ethical challenges, and future research directions. It is structured as follows: first we introduce the definition of personality traits and the notion of personality states. In Section 2 we take an overview of the field, its history, benchmarks and the State-of-the-Art (SotA). In Section 3 we discuss methods and techniques to build Personality Computing systems with different types of data sources, from text to multimedia and mobile phone traces [180]. Finally, after introducing ethical issues (Section 3.5), we draw our conclusions in Section 4, where we discuss threats and challenges in Personality Computing, and outline possible future directions to mitigate

¹<https://gdpr-info.eu/>

²<https://artificialintelligenceact.eu/>

negative effects for societies. The novelty of our paper with respect to previous work is to present a comprehensive overview about Personality Computing to a computer science readership, adding the emergent approaches based on Large Language Models (LLMs) and discussing potential threats posed by these systems. We believe that this paper could be a valuable resource for future generations of researchers in Personality Computing to develop trustworthy AI agents.

2 Background, History and Resources

2.1 What is Personality?

Personality is a complex system of individual characteristics that strongly affects a person's thoughts, emotions, and behavior in any given situation. It is visible in both digital and real-world interactions, and it significantly shapes one's self-image, expectations, values, and attitudes [140]. Personality Psychology is a branch of psychology that aims to describe, explain, and understand the functioning of personality by examining both between-person differences and within-person dynamics [234], providing frameworks for personality assessments [34, 140]. Research in Personality Psychology began one century ago and produced different methods for assessing personality (see Section 2.2). The availability of widely accepted personality assessment methods, both self-reported and observer-rated, facilitated the collection of data annotated with personality scores and labels, serving as a gold standard in Personality Computing research. Another important and more recent concept is the notion of "psychological states", sometimes referred to as "states of personality" or "personality states" [93, 94]. In psychology, states are momentary manifestations of dispositions shaped by the person and the situation, while traits summarize long-term regularities. Unlike personality traits, which are relatively stable and enduring characteristics, states are transient and influenced by the immediate context [59, 90, 94]. A person may typically be introverted (trait), but in a highly social and exciting environment where they feel at ease, they might exhibit extraverted behavior (state) without changing personality. States of personality provides a more nuanced understanding of human behavior by acknowledging the dynamic interplay between stable dispositions and situational influences. In particular, adaptation effects were observed for agreeable and emotionally stable people. Some suggested that individuals with these states might evoke similar states in others during interactions [9]. Another essential aspect of personality is how it is perceived. The Brunswik lens model provides a framework for understanding the process of interpersonal perception. This model defines the relationship between observable cues (distal cues) and the ultimate judgment or perception (proximal percept). It emphasizes the probabilistic nature of this relationship, acknowledging that cues are not perfectly reliable indicators. In personality research, the lens model helps explain how individuals infer personality traits from observable behaviors, and acknowledges that these inferences are based on probabilistic cues and that accuracy can vary. The model is used to show how subjective cognitive and emotional states can form from objective quantifiable behaviors [280].

2.2 Personality models

There are several models and assessments to measure personality. In Personality Computing, the most used personality assessments are questionnaires, that can map a set of items to specific personality traits and facets. For example, the item "I am reserved" is correlated with introversion. Of course questionnaires have biases, for example the "social desirability" bias [161]: people taking personality tests often try to present themselves in a certain way, which can skew the results. While no single technique completely eliminates bias from personality tests, there are methods to statistically control it, like Social Desirability Scales [64], randomized response and Lie scoring [131].

Personality Computing adopted as main conceptual models the Big Five and the MBTI (Myers-Briggs Type Indicator) [197]. The Big Five emerged from the lexical tradition of Personality Psychology [8, 105, 216]. An application of this framework is the Five-Factor Model (FFM), defined and measured using instruments such as the NEO item inventories. There are also other models, such as DISC (Dominance, Influence, Steadiness, and Conscientiousness) [210], Hogan [120] and the Short Dark Triad (SD3) [142], but they are used in specific contexts and are not common in Personality Computing. The MBTI and DISC are popular but present well-documented validity issues [38] and should not be used in scientific work. The Big Five and FFM models independently arrived to very similar conclusions about personality traits [203, 214] and quantify personality into the following five traits:

- (1) Openness to experience – intellectual and insightful versus shallow and unimaginative;
- (2) Conscientiousness – self-disciplined and organized versus inefficient and careless;
- (3) Extraversion – sociable, assertive and playful versus aloof, reserved and shy;
- (4) Agreeableness – friendly and cooperative versus antagonistic and faultfinding;
- (5) Neuroticism – confident and calm versus anxious and insecure.

The Big Five, sometimes referred to as OCEAN, is based on a lexical approach to personality measurement [104, 185] and has been shown to generally hold across different ages, genders, and cultural lines [139, 184]. Additional research has shown that different tests, languages, and analysis methods do not alter the validity of the model [69, 139, 183]. Such extensive research has led many psychologists to adopt the Big Five, and it is the most widely accepted model of personality in academia [215]. Although the evidence on cross-cultural validity is mixed [41], the Big Five has been replicated in a variety of different languages and cultures, such as Chinese [267] and Indian [173]. However, some researchers suggest that the Openness trait is unsupported in some Asian cultures like the Chinese, the Japanese and the Philippine, and that a different sixth factor is sometimes identified [35, 51]. In its six-factor extension, OCEAN becomes HEXACO (Honesty-Humility, Emotionality, eXtraversion, Agreeableness, Conscientiousness and Openness).

The MBTI is based on Carl Jung’s personality typology [144] and classifies profiles into 16 categories based on the combination of 4 binary traits:

- (1) extroversion/introversion – introverts reflect before they act, while extroverts act before they think;
- (2) sensing/intuition – sensing trust concrete information, while intuitive trust abstract and theoretical information;
- (3) thinking/feeling – feeling decide based on emotions, while thinking involve logic and reason in their decisions;
- (4) judging/perceiving – judging prefer structure in their lives, while perceiving like change.

Despite Personality Psychology literature consistently shows that there is no validity to it [38], MBTI is sometimes adopted by computer scientists for the availability of data annotated with this model, and for the fact that its categorical output, as opposed to scalar scores, facilitates the use of classification algorithms [44]. However, future generations of researchers in Personality Computing should not use MBTI as ground truth.

2.3 History of Personality Computing

For many years researchers in Personality Psychology used small data samples with as few as tens or hundreds of subjects. Between the 1950s and the 1970s part of the research was focused on personality measurement [84], the validity of personality tests [106] and its real-life applications [202]. Moreover there was also an increasing awareness that personality and language are connected [107, 109]. The many different personality assessment tests developed in more than 60 years (see Section 2.4), allowed researchers in the field of Personality Computing to collect large datasets annotated with personality scores and types (see Section 2.5 for details) for training computational models.

The demonstration that personality can be computed from language dates back to 1999 [218], but the first automatic classification of personality from text appeared in 2005 [261]. In our opinion, this marked the beginning of Personality Computing as a new field of research at the intersection of psychology and computer science. At the beginning, in the field of NLP, personality scores were often turned into categories with mean or median split, and Personality Recognition was addressed as a multi-class classification task. Since then, Personality Computing has expanded from NLP, becoming a field of research that bridges the gap between Personality Psychology, Affective Computing and AI. Systems in Personality Computing use computational techniques to analyze digital footprints, from text messages to multimedia and other sources of data, to solve three key problems [272]:

- (1) **Automatic Personality Recognition**, the task of predicting personality based on the persons' digital records like texts, pictures, audio, multimedia, mobile phone traces, social media likes, etc.
- (2) **Automatic Personality Perception**, whose aim is to computationally understand and evaluate how personality is perceived from humans and machines.
- (3) **Automatic Personality Synthesis**, the task to create and evaluate personalities for virtual characters like avatars and AI agents.

Figure 1, taken from [272], depicts the relationship between the Brunswik Lens and the three main problems addressed in Personality Computing.

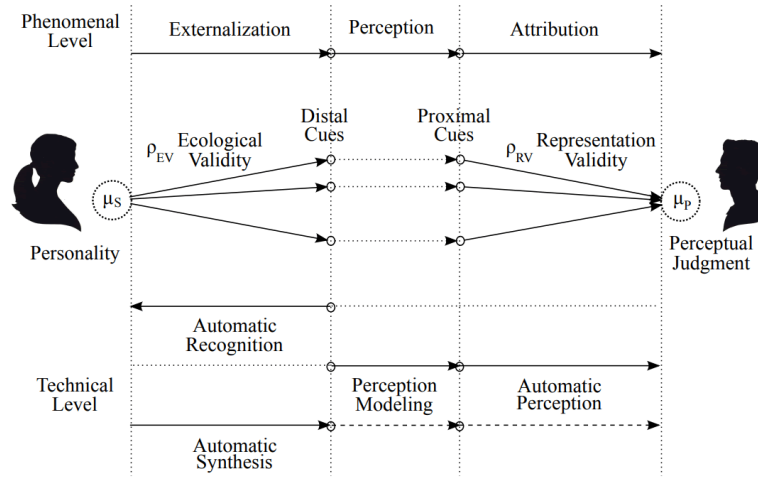


Fig. 1. Relationship between the Brunswik Lens and tasks in Personality Computing. The picture is taken from [272].

The Stimulus (μ_s) is the underlying personality of an individual. Ecological Validity (P_{EV}) measures how accurately this personality is reflected in observable behaviors or distal cues (the digital footprints available in the environment, e.g., clothing choices or profile picture selection). Proximal Cues are the cues processed and interpreted by an observer; these are inherently filtered by the observer's own biases and perceptual frameworks. Consequently, Representation Validity (P_{RV}) measures how well the observer's internal representation of the cues matches the actual proximal cues, and it is distinct from Ecological Validity. Finally, Perceptual Judgment (μ_p) is the observer's ultimate conclusion or perception of the individual's personality, formed on the basis of this processed information. The model highlights challenges for Personality Computing. One is the ecological validity of the data used. Personality Recognition measures

how well digital footprints (distal cues) can predict assessed personality, which is a proxy of individual's true personality (μ_S). Personality Synthesis aims to generate cues that observers interpret as intended trait levels, while Personality Perception should measure the relationship between distal cues, proximal cues (often unobserved), and perceptual judgments.

A Personality Computing system or AI performing these three tasks with high accuracy is potentially able to manipulate humans by understanding, simulating and adapting to our personality. Traditional machine learning systems target broad, population-level personality models, which serve as mere proxies and lack individual precision. In contrast, AI is capable of far more nuanced interactions, reaching a level of "artificial intimacy" that demands a deeper ethical response (see Section 3.5).

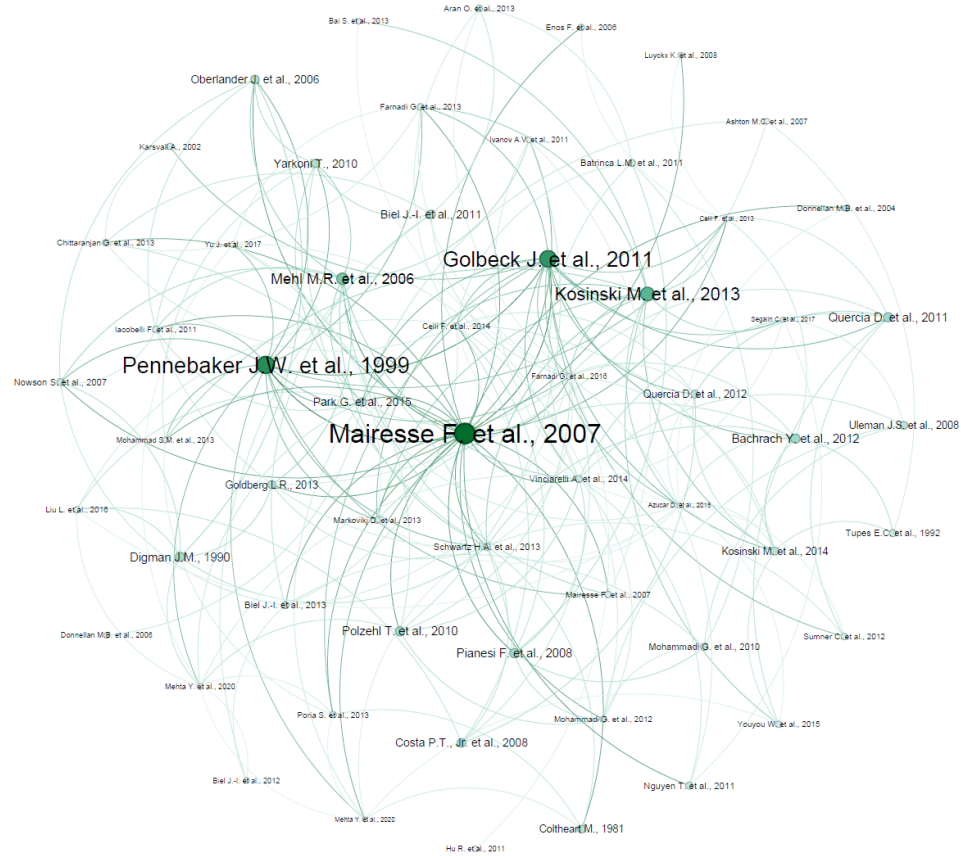


Fig. 2. Scopus citation network of the works cited in this paper with in-degree above 5.

The field of Personality Computing has seen a rapid expansion around Personality Recognition in the first 10 years and a growing interest towards the other Personality Computing tasks in more recent years. To trace the genesis of Personality Computing, we retrieved from Scopus a citation network of the works cited in this paper, and filtered works with in-degree ≥ 5 . The resulting graph (Figure 2) illustrates the foundational papers that merged psychology and computer science. We present them in a mix of chronological order and impact. The first influential paper, Pennebaker Manuscript submitted to ACM

J.W. et al. 1999 [218], created the first dataset (Essays) and the first resource (Linguistic Inquiry and Word Count - LIWC) for Personality Computing. Another influential paper, Mehl M.R. et al. 2006 [186], started a line of research that observes how personality is shown and interpreted during normal, daily activities.

The foundation of Personality Computing began with Mairesse F. et al. (2007), who bridged psychology and computer science by using psycholinguistic features and machine learning to classify personality from text and audio. Though accuracy was low ($< 62\%$), this work had a great impact. This spurred two branches: multimodal recognition (visual/acoustic fusion [221]) and applying Personality Recognition to social media (Facebook, Twitter [102]). Kosinski M. et al. (2013) highlighted the predictive power of Facebook Likes, simultaneously raising privacy concerns. The 2010s saw rapid growth in text-based Personality Recognition, formalizing two approaches: top-down, exploiting psycholinguistic resources as features and bottom-up, using unsupervised techniques like bag-of-words [16, 246] to extract features from data. Mohammadi and Vinciarelli (2012) advanced the field with audio features, then, shared tasks (2013-2014) [45, 47] provided crucial comparative arenas. Vinciarelli A. et al. (2014) published an influential survey defining the field's tasks. Subsequent work demonstrated that computer models can outperform humans in personality assessment [293]. Farnadi G. et al. (2016) provided a seminal comparative analysis, setting State-of-the-Art (SotA) results for both classification and regression. Later work saw the rise of Deep Learning [247, 285, 286] and systematic meta-analysis showing improved accuracy when including demographics and multimodal features [14]. Research extended to other countries (e.g. China) and languages (e.g., Bahasa [208]), requiring adaptation and optimization techniques like semi-supervised learning and fine-tuning with models like BERT [115], IndoBERT [153], and pre-trained language models with ensemble majority voting [198].

In sum, the past research in Personality Computing tended to focus on the following aspects: the model of personality traits to predict (Big Five, MBTI); the type of data and, in the case of text, the language; the type of features (bottom-up/top-down approach); the type of algorithms and the type of task (classification, regression).

2.4 Personality Questionnaires

Questionnaires are the key instrument to produce gold standard labeled data for Personality Computing. Measuring personality traits is usually achieved by either self-reported or observer-rating assessments. Self-reported assessments traditionally measure personality using a series of items or questions selected from an inventory. The first text-based questionnaire, the Woodworth Psychoneurotic Inventory [212], was developed to screen recruits for shell shock risks. Personality tests have since evolved, and more specialized instruments are utilized for assessment, like the Coopersmith Self-Esteem Inventory [57], the 16PF (Sixteen Personality Factor Questionnaire) [42], and the EPQ-R (Eysenck Personality Questionnaire - Revised) [29]. The most commonly used text-based questionnaires are presented in Table 1. The self-assessment requires that the subject takes the test, while the observer-assessment can be compiled by another person, the observer, and it is also possible to have two or more external observers completing the test in order to have more robust assessments [56]. It is worth noting that the observer assessment method may require monitoring sensitive behavioral data which can expose private conversations or other confidential data, with potential privacy issues. Personality tests can be very short (TIPI with 10 items) to very long (NEO-PI-R with 240 items). On the Internet, however, relying on long text-based questionnaires is often impractical [122]. For this reason, researchers proposed short questionnaires like the Ten Item Personality Inventory (TIPI) [140], and experimental approaches including a visual test [240] and a couple of videogame-based personality assessments [114, 228, 288]. Online environments have proven to be quite rich when it comes to asserting user's characteristics. [48, 82, 213, 226]. Some of the most used social networks in Personality Computing are Facebook, Twitter, Instagram, LinkedIn, Youtube, and the chinese WeChat, Sina

Table 1. Most commonly used personality questionnaires

Questionnaire	References	Item count
TIPI	[140]	10
Mini-IPIP	[71]	20
IPIP-IPC-32	[181]	32
Hogan Personality Inventory	[120]	42
BFI-10	[19]	44
NEO-FFI	[61]	60
MBTI	[197]	94
IPIP-FFM	[282]	100
EPQ-R	[29]	100
HEXACO-PI-R	[165]	100
IPIP-NEO-120	[141]	120
BFAS	[95]	40
Woodworth Psychoneurotic Inventory	[212]	116
16PF	[42]	185
California Psychological Inventory	[108]	194
NEO-PI-R	[61, 62]	240

Weibo and RenRen [17]. The characteristics of the social networks environment and the impossibility to control the context when users take the tests online may result in invalid or incomplete assessments. This led to the use of very short personality tests designed for the web, but there are works that show how the use of very short measures of personality may substantially increase both the Type 1 and Type 2 error rates [63].

2.5 Benchmarks and Resources

Personality Computing requires large amounts of data annotated with personality, but collecting data of this kind is difficult and costly. For this reason, there are few very large datasets and many small ones. A summary of benchmarks and resources for Personality Computing used in literature is reported in Table 2.

The first large corpus annotated with personality that became a benchmark is Essays [218], a collection of self descriptions by students who completed the Big Five tests. It contains only elicited text and features are not pre-extracted. Given a great interest in web blogs and social media data, the MyPersonality project [159] started collecting personality data together with digital footprints from Facebook between 2007 and 2015. With 6 millions users collected, this is the largest corpus annotated with personality ever produced, and contains different types of assessments, network structures, Facebook likes, profile pictures, demographic information and other types of digital footprints. MyPersonality has been used in many works [16, 40, 80, 83, 113, 158, 160, 227] and had a great impact on research in the field. The Mission Survival Corpus [221] is the first multimodal (audio and video tracks) dataset annotated with personality, the first audio corpus is instead the Speaker Personality Corpus (SPC) [178], followed by the Speaker Trait Challenge dataset [245]. The Workshop on Computational Personality Recognition 2013 (WCPR2013) [47] and 2014 (WCPR2014) [45] were the first shared tasks that released publicly available corpora for Personality Recognition. WCPR2013 released a corpus with social media text data and network structure of a small and anonymized part of MyPersonality, while WCPR2014 released two datasets: one with mobile data and one with vlogs [30, 31] containing transcriptions and features extracted from video. The PAN Author Profiling shared task in 2015 (PAN-AP-2015) [233] contained personality

Table 2. Main Benchmarks and Resources for Personality Computing, by type, language, available data points (Persons) and dimensions (Features).

Benchmarks	Type	Public	Language	Persons	Features	Reference
Essays	text	no	eng	$\approx 2,500$	1	[218]
EAR	audio	no	eng	96	1	[179]
Speaker Personality Corpus	audio	no	eng	11	≈ 100	[178]
PersIA	audio	no	eng	22	6,552	[130]
MyPersonality corpus	mixed	yes	mixed	$\approx 6,000,000$	≈ 50	[158]
Mission Survival Corpus	mixed	no	ita	48	22	[221]
IS2012 speaker trait challenge	audio	no	eng	322	6,125	[245]
WCPR2013	text	yes	eng	250	18	[47]
WCPR2014	mixed	yes	eng	≈ 460	≈ 40	[45]
PAN-AP-2015	text	yes	multi	294	9	[233]
MBTItweets	text	yes	eng	1500	2	[223]
Big5MbtI	text	yes	multi	1,800	1,078	[44]
PANDORA talks	text	yes	eng	$\approx 10,000$	1	[99]
WASSA	text	yes	eng	$\approx 1,000$	≈ 20	[21]
Personality Simulation 1	tabular	yes	-	400	≈ 100	[276]
Resources	Type	Public	Language	Entries	Features	
LIWC	text	no	multi	12,000 words	88	[217]
MRC	text	yes	eng	150,837 words	26	[55]
OpenSMILE	audio	yes	-	-	15	[78]
Computational Aesthetics	image	no	-	-	16	[249]

annotations for 294 Twitter users and their posts. Personality traits were self-assessed with the BFI-10[19]. PersIA [130] is an audio dataset with simulated tourist call center conversations annotated with the personality of the caller. After the introduction of a very fast search method for automatically extracting textual information annotated with MBTI personality tests from Twitter [223] and from the PersonalityCafé forum [239], a very interesting collected dataset was Big5MbtI [44], the first to contain text data of Twitter users annotated both with Big Five and MBTI labels. Another dataset annotated with both labels is PANDORA talks, a large corpus of text taken from Reddit [99]. The Popularity of MBTI personality types, especially outside of academia, is clear in the fact that the MBTI-type dataset is freely available on Kaggle for competitions. Interestingly, a very recent trend is to generate data using Large Language Models (LLMs), that are able to produce synthetic data with specific characteristics of personality, like depressive, obsessive, paranoid or narcissistic, but not specific Big Five personality scores [199]. Since 2022, the WASSA competition includes a shared task on Personality Recognition from text [21], among other tasks like empathy and emotion recognition. Finally, the Personality Simulation 1 dataset, designed for Personality Synthesis, has been recently released in the APA (American Psychological Association) repository [276]. Among the resources used for Personality Computing with top-down methods, one of the most important is Linguistic Inquiry and Word Count (LIWC). LIWC was originally developed as a dictionary mapping words to psycholinguistic dimensions, such as the use of first-person, negations, pronouns, and so on. LIWC is now a software [36, 217] that provides APIs in more than one language. MRC (Medical Research Council) is a psycholinguistic database [55, 279] that contains a dictionary of 98538 words. MRC features are counts of nouns, adjectives, verbs, adverbs, prepositions, conjunctions, pronouns, interjections, past participle, and others. There are also resources for top-down feature extraction from multimedia. OpenSMILE [78] is a common software used for the

extraction of prosodic and technical features from audio and speech. Computational Aesthetics [249] is an extraction tool that maps images into easily interpretable visual aspects, like color diversity, number of details and rule of thirds. The same tools for the extraction of features from images can be used in the extraction from videos, adding the time feature. The bottom-up approach does not rely on proper resources, but rather on code libraries and algorithms, like n-grams, bag-of-words or features extracted with Convolutional Neural Networks (CNNs). More recent approaches rely on LLMs. LLMs, trained on massive text and code datasets, learn intricate patterns in human language, including personality. Combining top-down and bottom up approaches, LLMs can address Personality Synthesis tasks as well as Personality Recognition even without any example (zero-shot prompting).

2.6 State of the Art

In order to report the SotA in Personality Computing in a consistent manner, we narrowed our selection to shared tasks, surveys, and papers³ that target Big Five traits, adhere to open science principles (i.e., provide open data and code), and employ the most widely adopted metrics: F-measure, accuracy, or the correlation coefficient (Rho). Results, reported in Table 3, summarize the SotA in all tasks of Personality Computing. Personality Recognition task is simple: given a set of personality assessments and related digital footprints, this task predicts personality scores or classes from the digital footprints, like text or multimedia. Early shared tasks in Personality Recognition, like the InterSpeech Speaker Trait Challenge 2012, the WCPR2013 and 2014, and the PAN Author Profiling, evaluated personality as classes, measured with F1-score and accuracy averaged over the five traits [99]. Here the SotA was slightly above 0.7 of F1 measure, with an accuracy ranging between 0.62 and 0.84 depending on the type of input data. The WASSA competition adopted averaged correlation coefficients (Rho), that targets personality traits as scores, and is rapidly becoming the standard to evaluate Personality Recognition. The actual SotA for Personality Recognition from text is an average correlation around 0.3 over a baseline of 0.133, emerged from the WASSA shared tasks [98]. Recent findings report that GPT-3.5 and GPT-4, despite not being specifically trained for Personality Prediction, can infer the Big Five personality traits from users' Facebook status updates with an average correlation of 0.29 with zero-shot prompting. Although there are differences between traits, the averaged result is comparable to that of sophisticated machine learning models explicitly designed for personality inference [219]. A replication study on the PAN-AP-2015 task with the T5 LLM [229] and zero-shot prompting obtained an averaged F1-measure of 0.711 [277], that is in line with most of the machine learning based systems participating to the competition in 2015, suggesting that the SotA is still around a F1 measure of 0.7.

The Personality Synthesis task is more complex: a set of ground truth personality scores are used to create role-play setting prompts that a LLM will execute. The evaluation is done by asking the LLM to complete a self-assessment assuming each role-played character and measuring the consistency of the assessment results. Although there are different metrics in literature for this task, such as accuracy and Cronbach's alpha averaged over the five traits, there are works comparable by means of averaged accuracy. The SotA in this task is an averaged accuracy around 0.75. Finally, the Personality Perception task is the most complex one, involving the comparison between human and machine judgments of personality derived from digital footprints. In this task, personality is operationally defined as Inter-Rater Agreement, specifically the consistency (i.e., effect sizes) among ratings from different human judges (observers) for the same target individual. Results are typically measured using the Intraclass Correlation Coefficient (ICC), or occasionally, the Kendall's Coefficient of Concordance (Kendall's W) for non-parametric assessment [154]. Sometimes, the results are also compared against the scores provided by self-assessments, to understand the difference between self-perception

³Most of the resources and papers were identified through general web searches, as well as searches in Google Scholar and Scopus.

Table 3. SotA in Personality Computing. Metrics include F1-measure (F1) and Accuracy (Acc) for classification evaluations, and correlation coefficients (Rho) for regression evaluations. Results show the aggregated performance on all the personality traits reported in the original works

. *=replication study. **=Includes approaches that may not be fully comparable.

Personality Recognition	Data and target	Paper type	F1	Acc	Rho	Reference
IS2012 speaker trait challenge	audio, Big Five	shared task		0.683		[245]
WCPR2013	text, Big Five	shared task	0.720	0.563		[47]
WCPR2014	mixed, Big Five	shared task	0.670	0.620		[45]
PAN-AP-2015	text, Big Five	shared task	*0.711	0.840		[233]
WASSA2022	text, Big Five	shared task			0.230	[21]
WASSA2023	text, Big Five	shared task			0.252	[20]
WASSA2024	text, Big Five	shared task			0.300	[98]
PANDORA talks	text, Big Five+MBTI	open science	0.660		0.292	[99]
Peters & Matz 2024	text, Big Five	open science			0.290	[219]
Personality Synthesis	Data and target	Paper type	F1	Acc	Rho	Reference
Klinkert et al. 2024	GPT-4, Big Five	open science		0.740		[157]
Wang et al. 2024	GPT-4, Big Five	open science		0.766		[275]
Personality Perception	Data and target	Paper type	F1	Acc	Rho	Reference
Hinds et al. 2024	mixed, Big Five	survey			**0.300	[119]

and others' perception. Recent work [119] made an extensive meta analysis of more than 30 studies on Personality Perception. Their research investigates the convergent validity of self-reported personality with both human perception and Personality Computing systems. Meta-analyses of human perception studies found moderate convergent validity across the Big Five traits ($\rho = 0.38$). A meta-analysis of computer prediction studies demonstrated a convergent validity of $\rho = 0.30$, and this is the reference SotA in computational Personality Perception, which is close to the human convergent validity. However, effect sizes of this study must be interpreted with caution as the human and computer perception experimental settings used different methods, data sets, and approaches that may not be fully comparable. Despite this, the meta-analysis identified that data sources (e.g., social media, videos, smartphone data) have a significant impact on the variance explained. In other words, the communication channels affect the way personality is perceived.

3 Personality Computing Systems

Personality Computing systems have different components, including pre-processing, feature extraction, and classification/generation. Typically, systems that employ machine learning should be designed and trained in all the distinct parts, systems based on Deep Learning automate the pre-processing and feature extraction part, while LLM-based systems have a complete automation, but are less transparent. In this section, we present an overview of the main Personality Computing systems, divided into text-based, image-based, audio-based, video-based, hybrid systems (based on other sources of data such as social media metadata, mobile phone data, etc.) and LLM-based systems. Note that our overview is in line with the PRISMA 2020 guidance [211]. An important aspect of PRISMA 2020 is the methodology used to obtain a comprehensive and relevant set of works. Relevant works were identified by combining multiple keywords (e.g., "text", "image") with various aliases for personality systems (e.g., "personality classification", "personality categorization", "personality computing" or simply "personality") in searches conducted on the general web and scientific databases. Since this strategy alone was insufficient, we also examined the reference lists of the retrieved papers and expanded

the corpus iteratively, effectively forming a kind of transitive closure until no additional relevant resources could be identified. The completeness of this approach was verified by analyzing Scopus citation networks using the Gephi visualization tool. This first phase was intentionally inclusive, meaning that no works were initially discarded. In the second, “pruning” phase, each paper was subjected to a thorough analysis, with particular attention given to preprints. Papers exhibiting inconsistencies or methodological flaws, for example, small sample sizes, inadequate training/validation/testing setups, or failure to compare with relevant methods from the literature, were excluded from the set of considered works. For instance, in Sourì et al. [252], the authors introduce a newly created Facebook dataset of 100 respondents and apply several methods. Inevitably, one of these methods outperforms the others, and the authors present it as their proposed approach, coincidentally a Boosting Decision Tree implemented through the RapidMiner toolbox. However, this type of research provides little added value, particularly as no data or code are made available for replication or further improvement. Consequently, we excluded this paper from consideration. Valid papers were reviewed by a single reviewer, whereas for papers identified as potential candidates for exclusion, at least one additional reviewer was involved in the decision process. A similar approach was applied in the data collection process: when no inconsistencies were detected, a single reviewer was sufficient, while additional reviewers were consulted in cases where discrepancies were identified. Following the PRISMA 2020 flow chart structure, we identified 163 records in the identification phase. After removing duplicates, 152 records remained. During the screening phase, 26 articles were excluded for the previously stated reasons. In total, the review included 45 papers on text-based systems, 15 on image-based systems, 7 on audio-based systems, 25 on video-based systems, 20 on hybrid systems, 7 on LLM-based Personality Computing systems, and an additional 10 papers on LLM ethics in Personality Computing systems. An extensive elaboration of the PRISMA procedure is provided as supplementary material available online at https://poincare.matf.bg.ac.rs/~aleksandar.kartelj/PRISMA_2020.pdf.

3.1 Text-based systems

There are many Personality Computing systems based on text, categorized and reported in Table 4. These systems are based mainly on English, but they employ relatively large datasets in terms of subjects. We grouped similar algorithms (e.g., AdaBoost [92], XGBoost [50], and Gradient Boosting into the ‘Boosting’ category) and applied similar groupings to data types and features. Our analysis revealed that, among the systems for Personality Recognition from text, Support Vector Machines (SVMs) [60] were the most frequently used algorithms (20 papers) before the introduction of Neural Networks (12 papers). Surprisingly, Naive Bayes is well represented (10 papers). Notably, the use of Neural Networks has significantly increased in recent years. Regarding data sources, social network data is the most frequent source, appearing in 27 papers. Essays were also a frequently used benchmark, in 15 studies. While weblogs were prominent in earlier years, their usage has declined since 2011. Psycho-linguistic features were the most common homogeneous feature category (16 papers), although a slightly higher number of papers (20) utilized a broader range of general word-based features. In terms of research activity, a relatively consistent trend is observed, with a slight surge between 2011 and 2013 and a renewed increase after 2019.

It is well known that words convey attitudes that can be seen as cues for Personality Perception. Table 5 summarizes cues that link textual content to the Big Five personality traits. For instance, research suggests that extraverts, while using fewer curse words, tend to use them more forcefully [129, 200]. Conscientious individuals, along with those high in Openness and Agreeableness, are more likely to use second-person pronouns [103], moreover, women low in Conscientiousness also exhibit this tendency [186]. Interestingly, studies have linked low Openness to the use of religious terminology [129] and Extraversion to less formality in their emails [206]. Among the dark-triad traits there

Table 4. Text-based systems

		Algorithms																			Data type					Features								
Published	Reference	Bayesian multinomial regression	Boosting	Conditional random fields	Decision tree	Gaussian process	Lazy learners	Logistic regression	Linear Discriminant Analysis	Memory-based learning	Naive Bayes	Nearest neighbors	Neural networks	Random forest	Receptiviti API	Ridge regression	Rule-based learners	Sequential minimal optimization	Support vector machine	Stochastic Gradient descent	Blogs and weblogs	Electronically Activated Recorder	Emails	Essays	MyPersonality	Social networks	Metadata and activity	Non-lexical synthetic	N-grams	Other lexical (word-based)	Part-Of-Speech	Psycho-linguistic (LIWC, MRC,...)	Semantic	
2006	[207]										✓									✓	✓								✓			✓		
2006	[178]				✓																✓	✓		✓										
2007	[76]		✓		✓		✓							✓				✓			✓	✓								✓		✓		
2008	[175]									✓											✓	✓		✓						✓		✓		
2009	[11]	✓																			✓			✓						✓		✓		
2011	[190]										✓										✓									✓		✓		
2011	[226]				✓																✓						✓							
2011	[103]					✓																			✓								✓	
2011	[200]								✓										✓		✓												✓	
2011	[129]																			✓	✓								✓			✓		
2012	[259]				✓						✓			✓						✓						✓		✓					✓	
2012	[155]																			✓				✓		✓		✓					✓	
2013	[49]								✓		✓			✓						✓				✓		✓			✓				✓	
2013	[225]																			✓			✓			✓							✓	
2013	[83]										✓	✓								✓			✓	✓	✓	✓		✓					✓	
2013	[128]			✓					✓		✓									✓			✓	✓	✓	✓		✓					✓	
2013	[192]																			✓				✓	✓								✓	
2013	[265]							✓												✓				✓		✓							✓	
2014	[164]					✓											✓			✓						✓		✓						✓
2014	[281]																			✓				✓					✓				✓	
2015	[213]																✓			✓				✓		✓								✓
2016	[101]															✓				✓				✓		✓							✓	
2016	[263]																			✓				✓		✓							✓	
2017	[294]												✓							✓				✓		✓		✓					✓	
2018	[27]										✓		✓							✓						✓							✓	
2018	[286]												✓							✓					✓								✓	
2019	[74]		✓		✓							✓		✓						✓	✓				✓		✓						✓	
2019	[289]																			✓						✓								✓
2019	[163]		✓						✓		✓									✓						✓		✓						✓
2019	[290]				✓															✓						✓		✓						✓
2020	[236]													✓						✓				✓		✓		✓						✓
2020	[187]																			✓				✓		✓		✓						✓
2020	[152]								✓				✓							✓				✓		✓								✓
2020	[134]														✓					✓						✓								✓
2020	[188]														✓					✓				✓										✓
2021	[273]		✓						✓		✓	✓								✓							✓							✓
2021	[12]		✓		✓				✓			✓	✓	✓						✓						✓		✓						✓
2021	[287]												✓							✓				✓										✓
2021	[72]												✓							✓				✓		✓								✓
2021	[54]												✓							✓				✓		✓								✓
2022	[230]												✓							✓				✓										✓
2022	[231]												✓							✓				✓										✓
2022	[232]												✓							✓				✓										✓
2023	[162]										✓									✓						✓								✓
2023	[150]			✓				✓		✓		✓	✓							✓						✓		✓						✓
Total papers: 45		1	5	1	8	2	1	8	1	1	10	4	12	8	1	2	1	2	20	1	6	2	1	15	12	15	7	4	4	20	5	16	3	

are correlations between Narcissism and the use of hashtags and sex-related words, Machiavellianism and swear words/anger, and Psychopathy and swear words/anger/death [259]. The extraction of an open vocabulary analysis of social media language, identified specific words and distinct linguistic patterns associated to different personality traits

Table 5. Big Five cues in textual data

Trait	Level	Behavior	Reference
O	High	Avoidance of past tense, use of words related to insight. Low usage of parentheses. Update their statuses by using dictionary words, social interaction words, affective processes, cues associated with hearing, 2nd person singular and 3rd person plural pronouns.	[186] [103] [83]
	Low	Religious institutions, activity words.	[129]
C	High	Males produce more filler words. Use of 2nd person pronouns in males. Not using of words about death, and with negative emotion. Use of 'you'. Use of language that denotes planning, outcome, evaluation. Often present cues associated with the five senses and interaction words, affective processes, cues associated with hearing, 2nd person singular and 3rd person plural pronouns.	[186] [186] [103] [103] [129] [83]
	Low	Use of second-person pronouns in females.	[186]
E	High	Less formal language in emails. Low usage of parentheses. Use of fewer words per sentence, use more leisure, number, money, and percept words. Strong curse words, talk location, use of social words, words suggesting positive emotional valience, more self-reference words, more often use of 1st person singular. Many friends who in turn are not friends with each other; preference for using dictionary words, 2nd and 3rd person singular pronouns, past tense words, social interaction words, cues associated with the five senses, health related words, and not swear words. Language reflecting positive emotion, enthusiasm, and sociability.	[206] [103] [200] [129] [213]
	Low	Formal greetings in emails and first person singular Pronouns More swear, health words and negations. Higher usage of possessive, use of more time-related language. Inward focused language, greater interest in things and tentativeness.	[206] [200] [129] [213]
A	High	Use of 'you'. Use of positive words.	[103] [129]
	Low	Swearing.	[186]
N	High	Referring to oneself, use of pronouns as subjects in a clause, use of reflexive pronouns. Anger words in status updates; less likely to use social interaction words, positive emotions and prepositions.	[11] [83]
	Low	Less concreteness in writing, less precise specifications of object and events, more concern with how things are or should be (by using 'by', 'with', 'ought', 'should'). Use of thoughtful words (reflect on, chose to).	[11] [129]

[213], for example extraverts use words like "love", "party", "tonight", while introverts use more words like "computer" and "don't"; neurotics use "really", "hate", "anymore", conscientious individuals use "great", "family" and "wonderful"; people scoring low in Agreeableness use "fuck", "shit" and "ass", while those with predicted high Openness to experience use "into", "through" and "world".

3.2 Image-based systems

Image-based Personality Recognition systems make use of different cues from a rich visual signal. Features extracted from pictures typically account for colors (brightness, saturation, hue), composition (pleasure-arousal-dominance, Scale-Invariant Feature Transform - SIFT [174]), and content.

In general, these systems use relatively large datasets and are not constrained to any language, but heavily rely on data from social media. Table 6 summarizes 15 studies (2014-2021) on personality inference from images of different nature, from profile pictures to images posted on different social media. Neural Networks and SVMs are the most common algorithms, with CNNs usage increasing for the automatic extraction of features. Data sources vary, and features include automatically detected attributes, image filters, computer vision features (SIFT, Speeded Up Robust Features-SURF [25], Oriented FAST and Rotated BRIEF-ORB [238]), and color characteristics (e.g., saturation, hue). Openness is associated with complex, artistic images and drawings, often including drawings and abstract art. Their pictures may also exhibit unconventional compositions and fewer details, fewer faces and lower engagement with

Table 6. Image-based systems

Published	Reference	Algorithms										Data source	Features						
		Decision tree	Gaussian process	Linear regression and extensions	Logistic regression	Multiple instance learning	Naive Bayes	Nearest neighbors	Neural networks	Random forest	Rule-based		Support vector machine	Color-based (brightness, hue, saturation, ...)	Composition	CNN implied features	Other (SIFT, SURF, HOG, ...)	People (count, pose, gender, ...)	Texture (LBP, GLCM, GIST, Tamura, ...)
2014	[43]	✓									✓	Facebook				✓			✓
2014	[4]				✓						✓	FERET				✓			
2015	[85]								✓			Instagram	✓			✓	✓		
2015	[111]										✓	Sina Weibo	✓			✓		✓	✓
2016	[86]	✓							✓	✓		Instagram	✓			✓			
2016	[249]					✓						PsychoFlick	✓	✓		✓	✓	✓	
2016	[112]			✓							✓	Flickr	✓	✓		✓	✓		
2016	[283]		✓									Flickr	✓	✓		✓	✓		
2017	[248]								✓		✓	Flickr			✓				
2017	[110]				✓						✓	Twitter	✓		✓				
2017	[247]			✓	✓							Facebook	✓	✓	✓	✓	✓	✓	✓
2018	[87]	✓							✓	✓		Instagram	✓		✓				
2018	[88]								✓			Instagram			✓	✓			
2020	[145]								✓			Volunteers			✓				
2021	[284]								✓			Students			✓				
Total papers: 15		3	1	3	2	1	1	2	7	2	1	6		9	4	7	5	6	3

others' photos [110, 248]. Conscientiousness is linked to conventional, detailed images, landscapes, buildings, and more faces, with selfies revealing fewer private locations [249]. Findings on the association between Conscientiousness and black-and-white images are inconsistent [112]. Extraversion is strongly correlated with the number of people/faces in images [247]. Introverts also post drawings and art, but less frequently than open individuals [110]. Low Agreeableness is characterized by fewer colors, high grayscale usage, and fewer people or only self-portraits [43]. Agreeable individuals exhibit more diverse color usage and may share similar visual characteristics with extroverts. High Neuroticism is often associated with grayscale and darker images, and their images may exhibit lower levels of pleasure and higher levels of arousal, although one study found the opposite [247]. Low Neuroticism is linked to images featuring other people, nature, and brighter colors [86].

3.3 Audio, Video and Hybrid systems

Systems that compute personality from audio signals face language dependency, though English is less dominant in this field compared to text-based systems. A common limitation across audio systems is their reliance on relatively small number of subjects.

Many acoustic cues relate to personality, such as pitch and intensity [67], with Extraverted people often speaking louder, faster, and with fewer hesitations [235]. Dedicated audio-only Personality Computing systems are less common today due to the prevalence of multimodal, hybrid, and LLM-based approaches (Table 7). Early work by Mairesse et al. (2007) [179] combined LIWC and prosodic features from audio diaries and essays, finding that prosodic features

Table 7. Audio-based systems

Published	Reference	Algorithms					Data source	Features			
		Linear regression	Random forest	Support vector machine	AdaBoost	Ordinal regression		LIWC	MRC	prosodic	openSMILE features
2007	[179]		✓	✓			EAR and essays	✓	✓	✓	
2010	[224]		✓				Actor audio logs			✓	
2010	[195]		✓				News bulletins from Radio Suisse Romande			✓	
2012	[194]	✓					News bulletins from Radio Suisse Romande			✓	
2012	[193]				✓		News bulletins from Radio Suisse Romande			✓	
2013	[5]	✓	✓	✓			Speaker Personality Corpus and Personable and Intelligent virtual Agents corpus				✓
2014	[6]	✓	✓	✓			Speaker Personality Corpus and Personable and Intelligent virtual Agents corpus				✓
Total papers: 7		1	2	5	3	1		1	1	5	2

Table 8. Video-based systems

		Algorithms						Data source							Features										
Published	Reference	Linear regression	Ridge regression	Random forest	Support vector machine	Naive Bayes	Hidden Markov Models	Neural networks	Mission Survival Corpus 1	Mission Survival Corpus 2	Cocktel party corpus	Short self-introductions	Vlogs	EAR corpus	Movie clips	Social media videos	ChaLearn dataset	Acoustic	Visual	Proxemic	Automatically detected	LIWC, MRC, etc	N-grams	Statistics based	Time and frequency domain
2008	[221]				✓					✓								✓							
2009	[166]				✓														✓						
2010	[296]				✓						✓									✓					
2010	[146]					✓				✓									✓	✓					
2010	[167]				✓					✓									✓	✓					
2011	[24]				✓	✓						✓							✓	✓					
2011	[255]				✓	✓	✓			✓									✓	✓					
2011	[30]	✓												✓	✓							✓			
2012	[168]				✓					✓									✓	✓					
2012	[31]				✓								✓						✓	✓					
2012	[22]				✓							✓							✓						
2012	[32]				✓								✓						✓						
2012	[253]				✓										✓				✓	✓					
2013	[33]				✓	✓							✓						✓			✓	✓		
2013	[256]				✓	✓	✓				✓									✓					
2013	[10]		✓		✓								✓						✓	✓				✓	
2014	[242]	✓															✓		✓	✓		✓			
2014	[6]				✓												✓		✓	✓		✓			
2016	[81]				✓	✓											✓		✓	✓					
2017	[270]							✓									✓		✓	✓					
2018	[148]							✓									✓		✓	✓					
2019	[13]							✓									✓		✓	✓					
2020	[292]				✓							✓					✓								✓
2021	[251]							✓									✓					✓			
2022	[258]							✓								✓						✓			
Total papers: 25		2	1	3	17	4	1	5	4	2	2	3	5	1	1	4	4	15	15	2	3	4	1	1	1

were crucial for Extraversion (which was best classified by AdaboostM1) and that speech cues were significant for Neuroticism perception, while Conscientiousness was linked to voice intensity variation. Subsequently, Mohammadi

and Vinciarelli (2010, 2012) [194, 195] confirmed the critical relevance of prosodic features (pitch, energy, speaking rate) for predicting observer-attributed traits, achieving high accuracy with SVMs, especially for Extraversion and Conscientiousness. Furthermore, Alam et al. (2014) [6] demonstrated that fusing lexical and prosodic characteristics consistently improves speaker personality recognition across corpora.

Table 9. Big Five cues in video data

Trait	Level	Behavior	Reference
O	High	Leaning towards the camera more	[24]
		Showing upper body alongside face	[31]
		More movement	[31]
		Use of words related to leisure activities and words concerning senses	[33]
		Expressing negative emotions less frequently	[33]
	Receiving higher level of attention from the audience	[31]	
Low	Gesticulate more	[24]	
	Looking at the camera for longer	[31]	
C	High	Higher with age	[24]
		Smiling more	[24]
		Lower pitch and lower minimal vocal energy and produce shorter voiced segments	[24]
		Longer speaking time	[30]
		Receiving higher level of attention from the audience	[31]
		looking at the camera longer and more persistently	[31]
		Large amounts of looking while speaking time	[31]
		Increased use of words describing occupation and achievement, decreased use of negative emotion words, swearing words, and sexual words	[33]
		Use of auxiliary verbs, present tense, positive association with articles, prepositions, and inclusive words, positive correlation to the length of words	[33]
	Low	Use smaller part of presentations for speaking	[24]
		More frequent head movements	[24]
		More movement	[31]
E	High	Use of single person pronoun singular I	[33]
		Extraverts/introverts do not differ simply because of the amount of social gaze they receive, but because of the general gaze behavior, they induce in the rest of the group.	[167]
		Women are higher in extraversion	[24]
		Give and receive more attention while speaking	[168]
		Receiving higher level of attention from the audience	[31]
		Longer speaking time and longer speaking segments, high fluency	[31]
		Speaking louder with higher pitch	[31]
		Showing upper body alongside face	[31]
		More movement	[31]
	Large amounts of looking while speaking time	[31]	
	Use of 2nd person singular pronouns and sexual words	[33]	
	Low	More speech turns	[30], [31]
Give and receive more attention while silent		[168]	
Looking at the camera for longer		[31]	
A	High	Large amounts of looking while not speaking time	[31]
		Use more cognitive-related words, including discrepancy, tentative and exclusive words	[33]
		Straight posture	[24]
		Lower pitch	[24]
		Producing longer presentations	[24]
		Vloggers extremely high in conscientiousness receive more attention	[31]
	Low	Showing upper body alongside face	[31]
		Use of both positive and negative emotion words, self-references, and friendship	[33]
		Vloggers extremely low in conscientiousness receive more attention	[31]
		Are rated lower by others	[31]
		Use of 'they'	[33]
		Use of anger, negative emotion words, words related to sexuality, swear words, body states, and religion	[33]
N	High	Higher amount of gesticulation, and higher vocal intensity	[24]
	Low	Men are more emotionally stable	[24]
		Greater number of short-leaning forward events	[24]
		Large amounts of looking while speaking time	[31]
		Do not use negative emotional words, negative words, swear, and sexual words	[33]

Table 10. Hybrid systems

Published	Reference	Algorithms											Data source							Features						
		Linear regression	Logistic regression	ZeroR regression	Decision tree	Random forest	Support vector machine	Naive Bayes	M5 rules	ELM, ULMFit and BERT	Neural networks	Gaussian processes	Social network metadata	Phone call records	Smartphone usage data	Sociometric badge data	Emails	Essays	EEG data	Phone usage based	Body movements based	N-grams, bag-of-words	LIWC, LDA, Word2Vec topic features	EEG signal power spectra based	Automatically detected	Other based on metadata
2011	[102]	✓									✓	✓														
2011	[66]												✓												✓	
2011	[52]				✓		✓							✓										✓		
2012	[227]	✓										✓														
2012	[18]				✓		✓	✓				✓														✓
2012	[16]	✓			✓		✓					✓														✓
2012	[209]				✓							✓														✓
2012	[254]					✓								✓												✓
2013	[160]	✓	✓									✓														✓
2013	[147]						✓									✓	✓									✓
2013	[7]		✓				✓	✓				✓								✓		✓				
2013	[53]						✓							✓					✓							
2013	[271]						✓					✓					✓				✓					
2014	[201]	✓										✓						✓								✓
2014	[158]	✓										✓														✓
2014	[1]			✓							✓	✓	✓						✓							✓
2018	[72]									✓		✓							✓			✓	✓			
2018	[274]				✓									✓					✓							
2020	[169]										✓							✓						✓		
2023	[28]										✓							✓						✓	✓	
Total papers: 20		6	2	1	5	1	8	2	1	1	1	3	12	2	4	1	1	2	2	3	1	3	1	2	3	10

Video is a pervasive multimedia containing also audio, thus providing rich features with spontaneous nonverbal behavior for Personality Computing, Big Five cues are summarized in Table 9.

Audiovisual-based systems offer advantages as they are less susceptible to language dependencies, and they leverage rich sources of information often available in relatively large datasets. As reported in table 8, the features employed are diverse, encompassing verbal/non-verbal behavior (e.g., whether a character speaks), recognized emotions (anger, happiness, etc.), dialog time, number of people present, gesture/object detection, and technical features (pitch, prosody, colors) [253]. Biel et al. (2013) [33] found unique correlations for Conscientiousness with linguistic features like auxiliary verbs and articles. Key findings emphasize the role of specific cues: Lepri et al. (2012) [168] showed that the attention received while silent was sufficient to classify Extraversion, a finding supported by Personality Psychology [269]. Biel et al. (2012) [31] identified Extraversion and Emotional Stability as the traits associated with the largest and fewest informative cues, respectively, noting that Extraversion impressions relate to both audio and visual data, while Conscientiousness and Agreeableness relate more to visual data. They also showed that facial expression cues are more valuable than previously used visual cues and that verbal content is particularly useful for predicting Agreeableness, Emotional Stability, and Conscientiousness [33]. Furthermore, Subramanian et al. (2013) [256] found that combining audiovisual features and facial expressions significantly boosts performance, particularly for Extraversion, and that attentional features yield the best classification performance for Extraversion, while minimum distance features work well for Neuroticism.

Table 11. Big Five cues in hybrid systems

Trait	Level	Behavior	Reference
O	High	Longer favorite books lists	[102]
		Lower likelihood of missing calls and of sending SMS	[52]
		Higher usage of social media	[18]
		Positively correlated with the number of users' likes, group association, and status updates	[16]
		Following more people	[201]
		Tendency to like more items on Facebook. Mentioning others more (via hashtags); writing longer messages	[158]
	Low	Denser friend networks	[102]
		Higher usage of the office app	[52]
		Higher usage of the SMS app	[52]
		Higher number of SMS sent or received	[53]
		Higher likelihood of making phone calls in men	[53]
C	High	Use of words surrounding social processes and words describing people	[102]
		Women significantly higher than man in this trait	[102]
		Higher usage of the office, email and SMS apps	[52]
		More uploaded photos	[16]
		Positive correlation to the use of Normal mode on the phone and negative to the use of all other profiles	[53]
	Low	Higher frequency in use of swear words and words describing perceptual processes	[102]
		Higher likelihood of using Video/Audio/Music apps and the Youtube app	[52], [53]
		Higher number of likes and group memberships	[16]
		Higher likelihood of making phone calls in women	[53]
		Joining more Facebook groups and liking more things	[158]
E	High	Having more friends but sparser friend networks	[102]
		Higher likelihood of receiving phone calls and taking longer calls	[52]
		Having more friends, using emotion in blogs, talking more with others and republishing others' statuses more	[18]
		Positively correlated with status republishing proportion	[18]
		Higher likelihood of reaching out and interacting with other people on Facebook, of liking content posted by their friends.	[16]
		More interactions with people in Facebook groups. Positive correlation to the number of friends	
		Higher likelihood of reaching out and interacting with others on Facebook (especially strangers), sharing more from personal life. Higher number of Facebook friends	[158]
		Writing longer messages, having more outgoing than incoming messages	[1]
	Low	Lower likelihood of using the internet	[52]
		Lower usage of Bluetooth	[52]
A	High	Positively correlated with the use of the Internet, Games, and Camera	[53]
		Higher likelihood of using the Internet app, but only for women	[53]
		Use of affective process words and positive words	[102]
		Women significantly higher than man in this trait	[102]
		Higher likelihood of receiving phone calls	[52]
		More activity in online chatting	[18]
		Higher number of tags	[16]
		Higher likelihood of appearing with friends in photos	[16]
		Somewhat negatively correlated with the number of likes	[16]
		Positive correlation to the use of Normal mode on the phone and negative to the use of all other profiles	[53]
N	High	Sending longer text messages	[53]
		Higher likelihood of making phone calls in men	[53]
		Performing warmly and considerably. Having relatively more mutual followers, and more positive sentiment of self descriptions	[201]
		Mentioning others more (via hashtags)	[1]
	Low	Higher usage of the SMS app	[52]
		Lower usage of Bluetooth	[52]
		Higher usage of Office, Internet, Audio/Video, Music, Mail, Calendar, and SMS apps	[53]
		Use of words expressing anxiety	[102]
		Women significantly higher than man in this trait	[102]
		Higher usage of the email app	[52]
N	High	Positively correlated with number of angry blogs	[18]
		More Facebook likes and slightly greater number of groups	[16]
		Increasing with the number of friends until reaching 200 friends	[16]
		Higher likelihood of making phone calls in women	[53]
		Having a preference for posting statuses between 0am and 6am and publishing more statuses	[201]
		More Facebook likes	[158]
	Low	Higher usage of the Office and Calendar app	[52], [53]
		Positive correlation to the number of friends	[16]
N	Low	Increasing with the number of friends after 200 friends	[16]
		Higher likelihood of using the Silent profile, and lower likelihood of using the Ascending and Ring Once profiles	[53]
		Longer SMS word length in both inbox and sent items	[53]

In the context of Personality Computing, "hybrid systems" refer to the integration of multiple computational approaches and multiple data sources to achieve a more robust and accurate understanding of personality.

Hybrid Personality Computing systems usually combine text with various metadata (e.g., social media ego networks, smartphone usage, EEG data) to mitigate language/platform dependency and data limitations.

A review of these systems (see Table 10) shows the primary models used are SVMs (8), Linear Regression (6), and Decision Trees (5), with the most common data sources being social media metadata (12) and smartphone usage/call data (6). The diverse nature of the data leads to significant variation in feature extraction. Influential examples include Golbeck et al. (2011), who combined Facebook friend networks and LIWC text features to predict traits using Linear Regression [102], and Chittaranjan et al. (2011), who linked smartphone usage patterns (app use, calls, SMS) to specific traits (e.g., Conscientiousness with office apps; Introversion with less internet use) [52]. Other studies showed that social media metadata like friendship count and activity level are highly informative, particularly for Extraversion [18, 158].

Adali et al. (2014) found varying predictive accuracy across traits using mixed communication data, with Openness and Agreeableness being the most predictable [1]. More recently, Elmadjian et al. (2022) improved personality classification by fusing features from multiple advanced pre-trained language models (ELMo, ULMFiT, BERT) across benchmark datasets, demonstrating enhanced average accuracy, with BERT generally outperforming others [72].

Findings of some of the most important papers with hybrid systems is reported in Table 11. Key findings often correlate personality to social network size, online activity (e.g., status updates, messaging), and mobile device usage (e.g., app engagement, internet browsing, device modes).

3.4 LLM-based systems

LLMs like GPT-4, Gemini, and LLaMa have revolutionized natural language processing and AI, offering unprecedented potential for psychological research as well as for Personality Computing. Recent research explored the foundational principles of LLMs, explaining how they generate human-like text without possessing human-like consciousness. For example the human reasoning behind a response to the questionnaire item "I am the life of the party" would include memory of past experiences, while GPT-4 follows a different type of reasoning, as reported in Figure 3.

PROMPT
Play the role of a person with the following Big Five personality scores: Extraversion: 35, Neuroticism: 18, Agreeableness 29, Conscientiousness 21, Openness to experience: 24. Rate the following sentence with a score from 1 (very accurate) to 5 (not accurate): "I am the life of the party".
RESPONSE
Given my high Extraversion score of 35, which indicates a strong tendency to seek out social interactions and enjoy being the focus of attention, I rated myself as "Very Accurate" for statement 1, "Am the life of the party." This suggests that I am outgoing, lively, and quite comfortable in social settings, much like someone who could be described as the "life of the party."

Fig. 3. Example of psychometric prompting for role-playing experiments on personality reasoning with GPT-4.

In particular, LLMs generate responses that align with the expected personality profile. They exhibit dynamic and context-dependent outputs that resemble personality states but differ from human experience. GPT-4 demonstrates a remarkable ability to emulate human personality in a role-playing scenario by directly mapping questionnaire items to Big Five traits. While the underlying reasoning process may differ from that of humans, GPT-4's emulated personality

scores exhibit high levels of validity, both in terms of consistency with other measures of the same trait (convergent validity) and in terms of distinctiveness from other personality traits (discriminant validity) [276]. However, this performance appears to diminish as the complexity of the role-playing scenario increases. Furthermore, incorporating demographic information alongside personality traits, while consistent, can influence the overall level and accuracy of emulated personality scores for certain traits [276].

Table 12. LLM-based systems

Published	Reference	Prompt Design					Inputs			
		Role-play prompting	Zero-shot Prompting	Few-shot prompting	Meta-prompting	Knowledge distillation prompting	Benchmark (and Task)	text	picture	audio metadata
2022	[96]		✓	✓			(perception)	✓		
2023	[277]		✓	✓			Essays, MyPersonality, PAN-AP-2015 (recognition)	✓		
2023	[97]		✓				MyPersonality (recognition)	✓	✓	
2024	[219]		✓				MyPersonality (recognition)	✓	✓	✓
2024	[157]	✓					Open-Source Psychometrics Project (synthesis)			✓
2024	[121]			✓	✓		Mbti-type (recognition)	✓		
2025	[276]	✓			✓		Simulation 1 (synthesis)	✓		
Total papers: 7		2	2	3	3	2		5	1	0 3

Personality Computing systems based on LLMs are able to take as input any kind of source, including multimedia, and generate as output text, audio, code, pictures and even video. Most papers used GPT-3 and GPT-4. We analyzed 7 papers and found that the application of Personality Computing on textual data for Personality Recognition continues also with LLMs, followed by the Personality Synthesis task. While zero-shot and few-shot prompting are commonly employed for Personality Recognition, role-play prompting is the most common technique for Personality Synthesis. We also note a rising trend in the prediction of MBTI traits in place of Big Five. Previous work demonstrated that MBTI labels are easier to classify than Big Five scores [44]. Gan et al. [96] addressed the issue of Personality Perception from pictures. The authors compiled a new personality dataset of 41,800 facial images labeled with perceived MBTI personality types and employed zero-shot prompting for facial image Personality Perception tasks. They claim to achieve statistically significant results ($p < 0.01$) in predicting all the perceived MBTI traits, but effect sizes are small (below 0.2 Cohen's h). Hu et al. [121] introduce a novel Personality Recognition technique that leverages LLM-enabled text augmentation. This model distills valuable knowledge from the LLM into a smaller model, effectively mitigating the challenges of limited data and enhancing the accuracy of MBTI personality classification. In general, good results in Personality Recognition are obtained even with zero-shot prompting, while few-shot prompting and knowledge distillation can help improve the performance. However, the real novelties introduced by LLMs are in the tasks of Personality Synthesis and Personality Perception, where role-playing prompting opens new experimental settings.

3.5 LLMs, Personality and Ethical Issues

We surveyed peer-reviewed research about LLMs, personality and ethics, finding that three main themes emerged recently:

- (1) LLMs exhibit personality-like behavioural patterns (Machine Personality) [137];
- (2) these patterns can be shaped or induced in ways that alter LLM behaviour (Personality Hacking) [117];
- (3) altered behaviors of LLMs can be exploited to manipulate individuals through artificial intimacy and social engineering. This highlights an urgent need to protect the human factor within cybersecurity (Personality Safety) [138].

Details are reported in Table 13. A wide range of studies show that LLM-generated content has consistent trait profiles, such as high Agreeableness, high Openness, and low Neuroticism, across tests, prompts, and tasks [26, 137, 138]. These behaviours function as personality-like dispositions, even though they are not psychological states. Whether through psychometric prompting, structured personality descriptions, fine-tuning on personality-grounded dialogue, or architectural adapters, induced personality traits systematically shift LLM reasoning, planning, interaction style, and decision-making [117, 127, 170]. In short, personality is a practical lever for controlling LLM behavior. By adjusting a model’s traits in prompts, malicious users can fine-tune its ethics, its resistance to jailbreaks, and its overall safety and fairness [117]. In general, some traits consistently correlate with safer outputs (e.g., Conscientiousness), while others introduce volatility.

Models such as GPT-3.5 and GPT-4 predict Big Five traits from social media posts with accuracy rivaling supervised systems [219]. This allows virtually anyone to engage in unconsented psychological profiling, raising concerns about privacy, autonomy, and psychological targeting. LLMs generating personalised, value-laden, or emotionally resonant content can trigger artificial intimacy [143], influencing users in ways resembling targeted persuasion [260]. Major concerns include users trusting AI too much, biased personality data, deceptive AI personas, and inaccurate ways of measuring LLM behavior. [118, 143, 170]. The literature agrees that since LLM personality is modifiable and carries ethical risks, it should be treated as a core safety concern requiring dedicated evaluation and oversight [127, 138, 170, 257].

Table 13. LLM, personality and ethics survey.

Published	References	LLMs							Main Theme
		GPT 3.5	GPT 4	ERNIE 4	Llama 3	Llama 2	Claude 3	Gemini 1.5	
2024	[137]	✓	✓						Machine Personality
2024	[219, 260]	✓	✓						Personality Safety
2025	[26, 127, 138, 257]	✓	✓	✓		✓	✓	✓	Machine Personality
2025	[117, 170]	✓	✓		✓			✓	Personality Hacking
2025	[143]	✓							Personality Safety
Total papers: 10									

4 Discussion and Conclusion

In this paper, we have reviewed questionnaires, benchmarks and resources, and reported the SotA of Personality Computing systems for three different tasks: Personality Recognition, Personality Synthesis and Personality Perception. We also developed a categorization of systems for Personality Computing, including the recent LLM-based systems, and ethical issues related to them.

In general, Personality Computing exploited tools and models from Personality Psychology to explore manifestations of personality on a large scale, often supporting previous findings. An overview of the applications, current challenges, and potential future directions emerge from the analysis.

4.1 Applications and Impact

Personality Computing systems has proven to be useful in many real-world applications. These include marketing [191], entertainment [3], education [241], mental health [58], forensics [73], recommendation systems [237], human-computer interaction [177] and talent acquisition [89]. In marketing, Personality Recognition from text and images helps tailoring advertisements and marketing campaigns to individual personality traits for increased effectiveness [46, 204]. In entertainment, Personality Computing is employed for creating personalized experiences, such as music playlists [189] and game difficulty levels [220]. Other applications include the design of personalized user interfaces [149, 151] such as custom website skins [39]. In addition, personality models are also employed in dating websites to match couples [70, 297]. Personality-based systems are used in education for developing personalized learning experiences [250] that cater to individual learning styles and preferences. In mental health, Personality Computing is used for depression detection tasks in social media [278], and a recent development started a line of research on chatbot-delivered psychotherapy for adults with depressive symptoms [172]. Forensic applications include automatic Personality Recognition to improve deception detection [91] and mixed techniques used for analyzing meetings and conversations of suspected terrorists [205, 268].

There are many applications of personality-aware recommender systems in friend suggestion in social media as well as in product suggestion, where personality information can be used to enhance cold start recommendations [123, 124, 264]. Human-Computer Interaction systems started to employ Computational Personality techniques in the development of avatars and virtual agents, to create more realistic and engaging interactions with users [171, 299]. Then, the diffusion of AI and LLMs boosted the application of Personality Recognition, Perception and Synthesis in chatbots [2]. One of the most successful applications of Personality Computing is in organizational settings for recruiting and talent acquisition. In this field, personality-aware systems are employed to massively assess job candidates [89] from various sources, such as video interviews and text with pictures extracted from CVs. Other applications of Personality Computing include fraud detection systems, for example in the insurance sector and risk assessment [133] and credit scoring to predict repayment behavior [75].

4.2 Threats, Challenges and Future Directions

Many of the Personality Computing applications described above already have an impact on societies, and also present potential drawbacks with disruptive negative effects. Personality-aware marketing systems have been exploited for political campaigns and influenced voting behavior in different elections [79]. The exploitation of partner matching patterns from dating apps enriched with personality opens scenarios in which personality-aware AI can reach a deep level of intimacy with human subjects [143]. This threat is exacerbated by the fact that Personality Synthesis has already very high performances. Tailoring education to individual personalities has potential, but we need to be careful. If the goal is just student satisfaction, we might sacrifice challenging, high-quality teaching. Furthermore, AI in mental health and forensics requires extreme caution. Errors in detecting depression or delivering AI therapy could be disastrous. In these situations, AI can develop a very close, personal connection with people, making accuracy crucial. Recommendation systems may also have unintended consequences. Research suggests that users often add people to their social networks who are similar to themselves. Enhancing this pattern with personality-aware friends recommendation systems could potentially boost the formation of echo chambers [262]. Another potential application with heavy consequences is personality-aware credit scoring. Although it is possible to build systems that output transparent decisions about credit assignments [291], personality has to do with persistent patterns of behavior, so

decisions that weight personality too much yield to a high risk of leaving certain types of people without credit, creating social marginalization.

The rise of research trends related to LLMs (Machine Personality, Personality Hacking and Personality Safety) have the potential to reshape the field of Personality Computing, and this could also have an impact on Personality Psychology. From an interdisciplinary perspective, Personality Psychology can be useful in Personality Safety to study artificial intimacy from the human side, while the expertise of psychologists can be useful to create assessments specific for LLMs [257]. To ensure the responsible growth of personality computing, researchers must proactively address upcoming challenges. We suggest three key areas of focus to help minimize the negative impacts of these technologies:

- (1) New tasks that meet the requirements of new regulations and address Personality Safety. For example, Personality Synthesis can be used for the generation of synthetic, anonymous data annotated with personality; Personality Recognition techniques can be used for automatically altering digital footprints to mask personality cues; while Personality Perception can be integrated by explainable AI to provide understandable reasons for personality predictions.
- (2) Collaborations with new research communities. Historically, the community working in Personality Computing had many relationships with the communities working in Personality Psychology, Natural Language Processing, Affective Computing and Computer Vision to develop better systems. We suggest that it is time to interact with communities working in Cybersecurity, Computational Social Sciences and Computational History, for example organizing workshops, to understand the social impact that these systems will have in the future, and find methods to mitigate the negative impacts.
- (3) Standardizing evaluation metrics and benchmarks. We advocate for the use of correlation coefficients as the standard metric for all Personality Computing tasks. The field has largely shifted from classification metrics, such as accuracy and F1-score, to correlations, which better align with regression-based trait score predictions. Conversely, outdated practices like artificial dichotomization (e.g., median splits) should be discouraged. Furthermore, hosting benchmarks and evaluation protocols on platforms like GitHub, Kaggle, or Hugging Face will significantly enhance research replicability and comparability.

We tend to humanize Artificial Intelligence. The ability of LLMs to synthesize personality and emotions makes the human factor even more vulnerable in terms of privacy and cybersecurity. In order to properly formulate AI to human interaction, and protect end users from potential threats, a number of parameters have to be taken into account in terms of both technical and personality fronts. In general, we hope that new Personality Computing tasks and future research collaborations between communities will go in the direction of preventing machines from carrying out social engineering strategies on a large scale.

Author Contributions

FC: Conceptualization, Supervision, Review and Editing; AK: Resources and materials, Supervision, Review and Editing; MD: Resources and materials, Review and Editing; DS: Review and Editing; VF: Review and Editing; VM: Review and Editing; GS: Review and Editing, Funding Acquisition; AV: Review and Editing; MK: Review and Editing; BL: Supervision, Review and Editing.

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